

DETERMINISTIC PATH ENSEMBLES, DRIVEN COLLISION MAPS, AND ENDOGENOUS CONJUGATE PARAMETERS

QIAN QI

ABSTRACT. We separate the physical two-dimensional Liouville section from its Cantor symbolic natural extension. The generalized baker map is realized as an autonomous Hamiltonian impact suspension on the regular physical section, while all generated seam currents are controlled on a directional two-sided anisotropic scale. This yields finite-order physical response and a conditional canonical-cocycle calibration without treating symbolic endpoint germs as smooth phase points.

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1. ROUND-NINE RECONSTRUCTION: A PHYSICAL IMPACT SECTION AND A DIRECTIONAL CURRENT CALCULUS

The physical Poincaré section and the symbolic natural extension are different objects. The former is a two-dimensional symplectic section with a finite one-step singularity set; the latter is a Cantor path space used to resolve all iterated one-sided limits. No symbolic graph is called a Liouville section in this revision.

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1.1. The regular physical section and its path factor. Let $p_i(a) > 0$, $\sum_{i=1}^d p_i(a) = 1$, and $s_i(a) = \sum_{j \leq i} p_j(a)$. On $\Sigma = (0, 1)^2$ put

$$I_i(a) = (s_{i-1}(a), s_i(a)), \quad B_a(q, p) = \left(\frac{q - s_{i-1}(a)}{p_i(a)}, s_{i-1}(a) + p_i(a)p \right), \quad q \in I_i(a).$$

The one-step seam set is $\mathcal{S}_a = \bigcup_i \partial I_i(a) \times (0, 1)$. Let

$$X_a = \Sigma \setminus \bigcup_{n \in \mathbb{Z}} B_a^{-n} \mathcal{S}_a.$$

It has full Lebesgue measure and is invariant. Put $\Omega = \{1, \dots, d\}^{\mathbb{Z}}$ and let ν_a be the Bernoulli law with weights $p(a)$.

Theorem 1.1 (Physical-symbolic separation and exact factor). *The restriction $B_a : X_a \rightarrow X_a$ is an invertible measure-preserving map. The itinerary map*

$$\mathbf{c}_a : X_a \longrightarrow \Omega, \quad (\mathbf{c}_a x)_n = i \iff B_a^n x \in I_i(a) \times (0, 1),$$

is a bimeasurable isomorphism modulo null sets and $(\mathbf{c}_a)_\# m = \nu_a$. Endpoint expansions belong only to the symbolic completion; they are not extra points of the physical symplectic section.

Proof. Each branch has Jacobian one and sends a vertical strip bijectively to the corresponding horizontal strip. Removing all forward and backward seam preimages makes every iterate and inverse iterate single valued. A cylinder $[i_{-r} \dots i_s]$ pulls back to a rectangle of area $\prod_{k=-r}^s p_{i_k}(a)$. Cylinders generate the two sigma fields, and their diameters tend to zero in both stable and unstable directions. Hence the itinerary is one-to-one and onto modulo endpoint sequences and carries Lebesgue measure to ν_a . $\square$

1.2. An exact Hamiltonian impact suspension. For branch i let

$$S_i^a(q, P) = \frac{(q - s_{i-1})(P - s_{i-1})}{p_i}.$$

Then $Q = \partial_P S_i^a$ and $p = \partial_q S_i^a$, so the branch reset is exact symplectic. Consider the extended phase space

$$\mathcal{M}_a = \Sigma \times \mathbb{T}_\tau \times \mathbb{R}_E$$

with symplectic form $dq \wedge dp + d\tau \wedge dE$. Between impacts use $H = E$. At $\tau = 1$ apply the canonical reset generated by S_i^a when the incoming point lies in $I_i(a) \times (0, 1)$, and reset τ to zero. The incoming side determines the reset at a wall; the wall itself has zero Liouville flux.

Theorem 1.2 (Autonomous Hamiltonian impact realization). *The preceding data define a piecewise smooth autonomous Hamiltonian impact flow whose first-return time to $\{\tau = 0\}$ is one and whose return map is B_a on the full-measure invariant set X_a . The reset preserves the section form $dq \wedge dp$ and the induced Liouville flux. The construction is C^{m+1} in a on every compact parameter interval on which $\min_i p_i(a) > 0$.*

Proof. Hamilton's equations give $\dot{\tau} = 1$, $\dot{q} = \dot{p} = 0$ between impacts. At the impact the generating-function identities give exactly the displayed branch formula. The determinant is one and the canonical one-form changes by dS_i^a , hence the reset is exact symplectic. Branch interiors are disjoint, so the impact rule is deterministic. Seam points have zero incoming flux and are removed in X_a . Smooth parameter dependence follows from the explicit formula and the uniform lower bound on the branch widths. $\square$

This is a Hamiltonian impact system, not a claim that the Cantor natural extension is a smooth section. A smooth regularization is obtained by replacing the instantaneous reset by a thin canonical layer generated by S_i^a on each incoming port; its scattering map converges in every fixed C^r norm away from the seam.

1.3. A directional two-sided anisotropic scale. Let $\mathcal{F}_{r,s}$ be the sigma field generated by $\omega_{-r}, \dots, \omega_s$. Write $\Delta_{r,s}$ for the double martingale difference

$$\Delta_{r,s}f = E(f \mid \mathcal{F}_{r,s}) - E(f \mid \mathcal{F}_{r-1,s}) - E(f \mid \mathcal{F}_{r,s-1}) + E(f \mid \mathcal{F}_{r-1,s-1}),$$

with the obvious one-sided conventions. Fix $0 < \beta < \alpha < 1$ and $0 < \beta_- < \alpha_- < 1$ with $\beta/\alpha < 1$. Define

$$\begin{aligned} \|f\|_+ &= |\nu_a(f)| + \sum_{r,s \geq 0} \alpha^{-r} \beta^{-s} \|\Delta_{r,s}f\|_{L^1(\nu_a)}, \\ \|f\|_- &= |\nu_a(f)| + \sum_{r,s \geq 0} \alpha_-^{-r} \beta_-^{-s} \|\Delta_{r,s}f\|_{L^1(\nu_a)}. \end{aligned}$$

Choose the weak parameters so that $\alpha_-^{-1} < \alpha^{-1}$ and $\beta_-^{-1} < \beta^{-1}$. Let $\mathcal{B}_a^+ \hookrightarrow \mathcal{B}_a^-$ be the corresponding completions. The transfer operator is $\mathcal{L}_a f = f \circ \sigma^{-1}$.

Lemma 1.3 (Directional Lasota–Yorke estimate). *There are $C < \infty$ and $\rho = \beta/\alpha < 1$ such that*

$$\|\mathcal{L}_a^n f\|_+ \leq C \rho^n \|f\|_+ + C \|f\|_-, \quad n \geq 0.$$

The inclusion $\mathcal{B}_a^+ \hookrightarrow \mathcal{B}_a^-$ is compact.

Proof. Composition with σ^{-1} sends a block with past/future depths (r, s) to $(r+1, s-1)$ while $s \geq 1$. Its strong weight is therefore multiplied by β/α . After s steps the block has no future coordinate; those terms are bounded by the weak norm. Splitting the martingale sum into $s \geq n$ and $s < n$ gives the estimate. The translated-cylinder counterexample is absorbed because a cylinder at future depth N has initial strong weight β^{-N} . For compactness, truncate (r, s) to a finite rectangle; the truncation is finite dimensional and the remaining tail is bounded by the strict ratios α/α_- and β/β_- . $\square$

Theorem 1.4 (Spectrum on the directional scale). *The eigenvalue one is simple on $\mathcal{B}_a^+$ and the remainder of the spectrum lies in a disk of radius strictly smaller than one. The reduced resolvent*

$$\mathcal{R}_a = (I - \mathcal{L}_a)^{-1}(I - \Pi_a)$$

is bounded from $\mathcal{B}_a^+$ to itself and from each order of the parameter-current scale below to the next weaker order.

Proof. Lemma 1.3 gives quasicompactness. If $\mathcal{L}_a f = e^{it} f$ with $|e^{it}| = 1$, equality in the martingale contraction forces every future martingale difference to vanish. Applying the same argument to the inverse on the dual scale removes every past difference. Thus f is constant and $e^{it} = 1$. The resolvent statement follows from the spectral decomposition and the same strong/weak estimate. $\square$

1.4. All-depth physical currents and response. For a finite future word $w = i_0 \dots i_n$, the corresponding physical cut is an affine vertical line whose coefficient is a product of branch widths. Let $[\partial w]$ be its oriented line current. For $k \leq m$ define

$$\|J\|_{m,\text{cur}} = \sum_{k \leq m} \sum_w (1 + |w|)^m \nu_a([w])^\gamma |J_k(w)|, \quad 0 < \gamma < 1.$$

On a compact parameter interval, $\nu_a([w]) \leq (1 - c)^{|w|}$, so the current series and all finite parameter orders are summable. The coding map is used only to index these physical line currents.

Lemma 1.5 (Material derivative on every generated cut). *For each finite order m , differentiation of the physical transfer operator has the form*

$$D_a \mathcal{L}_a = (D_a \mathcal{L}_a)_{\text{bulk}} + \sum_w c_w(a) [\partial w],$$

where the coefficient and all derivatives through order $m - 1$ satisfy the current norm above. The same statement holds for iterated derivatives, with corners represented by products of compatible one-dimensional currents.

Proof. Differentiate the branch formula on each finite cylinder before taking the limit. The derivative of an indicator is the oriented boundary current and the derivative of its position is a sum of at most $|w|$ logarithmic branch width derivatives. Thus the order- k coefficient is bounded by $C_k(1 + |w|)^k \nu_a([w])$. Exponential cylinder decay dominates every polynomial and proves normal convergence. Compatibility under refinement is exactly finite additivity of oriented currents, so the completion is Hausdorff. $\square$

Theorem 1.6 (Finite-order physical response). *Let G_a be a physical observable whose pullback to Ω and whose shape currents have m finite orders in the preceding scales. Then $a \mapsto \int_{X_a} G_a dm$ is C^m . Every derivative is a finite sum of normally convergent words in*

$$D_a^j \mathcal{L}_a, \quad \mathcal{R}_a, \quad D_a^j G_a,$$

and every seam contribution is the pushforward of an explicitly indexed physical current. Common-root finite differences converge to the same response array.

Proof. Differentiate the eigenrelation $\nu_a \mathcal{L}_a = \nu_a$ on the directional scale and solve the centered part with $\mathcal{R}_a$. Lemma 1.5 supplies the shape derivatives and their normal summability. Iteration gives the usual finite Kato words with one strong-order loss at each current insertion. Under the common inverse-CDF root coupling, the only discontinuities in the parameter are the same moving cylinder boundaries; dominated convergence away from them and the current formula on them identify the finite-difference limit with the operator derivative. $\square$

1.5. Work and conditional calibration. Put a one-form $\eta = c_i d\tau$ on the complete incoming port of branch i . The work in n returns is $A_n = \sum_{j < n} c_{\omega_j}$. The canonical tilt is therefore the exact path-law change

$$\left. \frac{d\nu_{a,\theta}}{d\nu_a} \right|_{\mathcal{F}_n} = \exp\{\theta A_n - n\Lambda_a(\theta)\}.$$

Theorem 1.7 (Mechanical work and conditional coefficient identification). *The impact suspension accumulates exactly A_n and its tilted path law is the Bernoulli exponential family above. A cash-additive, time-consistent certainty equivalent has the same coefficient as the mechanical conjugate field precisely when its likelihood cocycle is required to equal this canonical mechanical cocycle. No identification is asserted for an unrelated preference family.*

Proof. Each flight crosses exactly one incoming port and the integral of $c_i d\tau$ over that unit flight is c_i . The cylinder Radon–Nikodym derivative is the displayed exponential. Exponential-utility rigidity gives a constant certainty-equivalent coefficient; equality of the two nonconstant likelihood cocycles forces equality of their exponents. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Every cross-paper input is typed in the round-nine dependency ledger. Repository proof and build gates are internal checks and do not replace independent external mathematical review.

PEKING UNIVERSITY, BEIJING 100871, CHINA
Email address: qiqian@pku.edu.cn